Universal gravitational energy

antocciphysics | TEACHER KEY | CPA

Learning target: Use the zero-at-infinity convention for gravitational potential energy.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 6; 3 course days later.*
No. Work and kinetic-energy change must refer to the same endpoints. Measure matching intervals.

02. Explain a worked example (worked)
-40 to -20 moves toward zero; outward motion increases U.

03. Look and explain (model)
Less negative: U magnitude halves.

04. Build the solution (practice)
-8 J;-4 J;ΔU+4 J.

05. Use the idea (reason)
Potential zero is convention; kinetic energy½mv² is non negative. Total can be negative for bound motion.

06. Choose a method (mixed) *Review: Unit 3, Day 11; 12 course days later.*
Shift becomes 4 times as large: shift=Δv×time. The toy model omits orbital curvature/gravity and uncertainties.

07. Exit check (exit)
+4 J; gained.

08. STRETCH: Apply it another way (stretch)
-30 and -10 J; outward change +20 J.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 6; 3 course days later.*
W=triangle area=1.5 J; $v_f \approx 3.162$ m/s.

02. Explain a worked example (worked)
-40 to -20 moves toward zero; outward motion increases U.

03. Represent and interpret (model)
Less negative: U magnitude halves.

04. Solve and justify (practice)
$U_i = -5.714 \times 10^{10}$ J; $U_f = -2.857 \times 10^{10}$ J;$\Delta U = +2.857 \times 10^{10}$ J.

05. Analyze the reasoning (reason)
Potential zero is convention; kinetic energy$\frac{1}{2}mv^2$ is non negative. Total can be negative for bound motion.

06. Choose a method (mixed) *Review: Unit 3, Day 11; 12 course days later.*
$\Delta v \approx 0.0006667$ m/s; impulse$\approx 5.333 \times 10^5$ N·s. Uses constant added transverse velocity over a straight-line comparison.

07. Exit check (exit)
$U_i = -12$, $U_f = -4$;$\Delta U + 8$ J; gravity work -8 J.

08. STRETCH: Apply it another way (stretch)
$\Delta U = (2/3)\ GMm/R$, gravity work negative of this; reaching infinity requires GMm/R total increase from R.